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# Building an In-House Macro Library for C-QTc Analysis

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## ABSTRACT

Early oncology trials increasingly use concentration–QTc (C-QTc) analyses to link drug exposure with QT/QTc changes and assess cardiac risk. While ICH E14 provides guidance, the study Modeling and Simulation Analysis Plan (MAP) defines the assumptions and models used for the analysis. Producing tables, listings, and figures (TLFs) also requires a robust and standardized framework. This paper outlines the design principles for an in-house SAS macro library for C-QTc analyses. The same principles can also be applied to R functions or packages. The objective is to support ICH E14 compliance and streamline consistent TLF production. The macro library covers assumption check, model selection and evaluation, and prespecified secondary analysis.

## INTRODUCTION

Certain drugs may unintentionally prolong the QT interval in patients. A prolonged QT interval can be a warning sign of proarrhythmic risk and may increase the risk of Torsade de Pointes (TDP), which can lead to fainting or sudden cardiac arrest. For this reason, concentration–QTc (C-QTc) analyses are important for linking drug exposure to QT/QTc changes and assessing cardiac risk in early-phase studies of new drugs across all therapeutic areas, including oncology.

Several papers and conference proceedings have discussed the challenges and solutions related to preparing analysis data and generating tables and plots for C-QTc analysis. Parkinson et al. (2025) provides a tutorial with practical steps for preparing an analysis-ready dataset, performing exploratory data analysis to check model assumptions, and explaining the linear mixed-effects model. Their workflow includes sample plots and tables, as well as R code for the model. On the SAS side, Sun (2020) and Zhang et al. (2025) describe the ADaM dataset for C-QTc analysis and the statistical analysis process, including SAS code and sample plots.

Although these publications provide good coverage of the tables and plots used in C-QTc statistical analysis, they do not discuss the code repository structure, such as a SAS macro library or R package. Therefore, this paper focuses on how to build a good SAS macro library for C-QTc analysis. The examples use SAS code, but the guidance below also applies to R function and package development.

### ICH E14

To provide guidance for assessing that risk early in drug development, in 2005 the International Council for Harmonisation (ICH) published the guideline document, ICH E14. This is the original guidance establishing the clinical approach to assess QT/QTc prolongation risk for non-antiarrhythmic drugs. There are revisions done after that. In 2017, ICH E14 updates with questions and answers guidance for industry.

### MAP (MODELING & SIMULATION ANALYSIS PLAN)

To work on C-QTc analysis in line with ICH E14, clinical study sponsors combine ECG data, PK concentration data, study medication data and conduct the assumption checking and modeling to evaluate whether the treatment drugs have impact on prolonging QT interval.

As the key guidance and planning document for the C-QTc analysis, the Modeling & Simulation Analysis Plan specifies data, models, covariates, treatment drugs, and mapped to the high-level guidance provided from ICH E14 and Scientific White paper (2018) on concentration-QTc modeling. The MAP has four

major sections: Assumption Checking, Model Evaluation, Model-Based Prediction, and Secondary Analysis.

**Assumption Checking**

- No drug effect on HR
- QTc interval is independent of HR
- No time delay between drug concentrations and ΔQTc
- Linear C-QTc Relationship

**Model Evaluation**

- Scatter plot of individual model predicted vs. observed ΔQTcF, overlaid with loess smooth line and 90% confidence band.
- Residuals will be studentized and the following plots will be generated:
    - Quantile-quantile plot of studentized residuals to check for normality.
    - Studentized residuals versus concentration, overlaid with loess smooth line and 90% confidence band.
    - Studentized residuals versus time, overlaid with loess smooth line and 90% confidence band.
    - Studentized residuals versus baseline QTcF, overlaid with loess smooth line and 90% confidence band.
- C-QTc quantile plots will be provided.

**Model-Based Predictions**

- The estimated mean QTcF change from baseline (ΔQTcF) and 90% confidence interval will be obtained from the model at the observed geometric mean Cmax for each relevant day/dose evaluated in the study. The model may also be used to predict the mean effect on QTc at other exposure levels of interest.

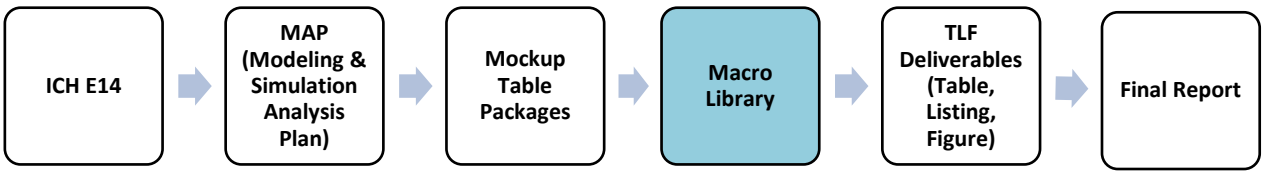
**Secondary Analyses**

- Descriptive Analysis
- Categorical Analysis

## MOCKUP TABLE PACKAGES

These checking and analysis are carried out by several tables and figures. A mockup table package is provided from statistician to programming team to follow the guidance and create tables, listings, and figures.

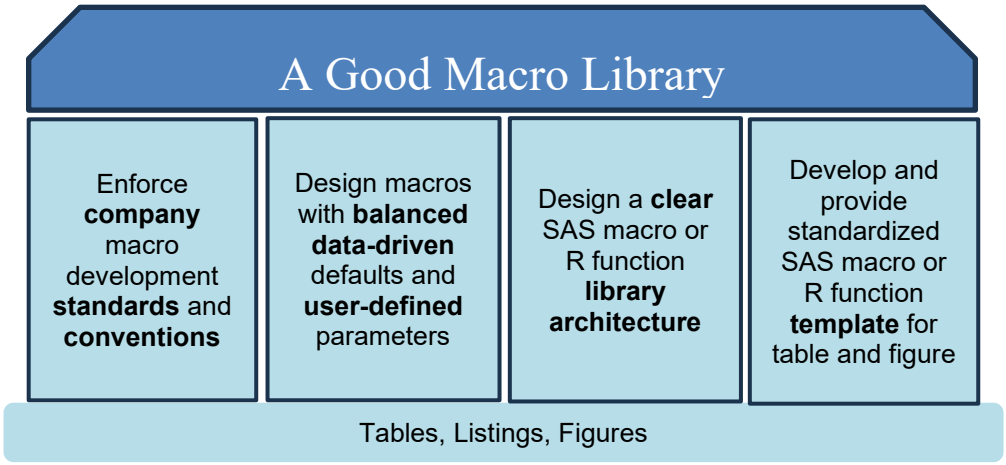
## FLOW FROM GUIDANCE DOCUMENT TO DELIVERABLES




## GUIDING PRINCIPLE OF A GOOD MACRO LIBRARY

SAS macros and R functions for tables, listings, figures reduce repetitive effort in the coding. An ideal in-house SAS macro library or R package follows company standard convention, flexibility of usage, clear roadmap for different franchise, indication, protocol with design. A well-designed macro library or R package allows different studies to easily adopt and utilize it. The importance of a well-designed macro architecture has been recognized in Redner & Coughlin (2007).

This paper outlines the guiding principles of a well-designed macro library:




**Display 1: guiding principle of a good macro library**

## ENFORCE COMPANY MACRO DEVELOPMENT STANDARDS AND CONVENTIONS

- Use consistent macro parameter names
  Companies usually have standard naming rules for macro parameters. A typical macro may include parameters for the input dataset, data selection criteria, treatment arm variable, output file name, output folder location, page orientation, page margins, table body font size, subtitle, data cutoff date, data citation text, and debug option. For these commonly used macro parameters, consistent naming makes the macro easier to understand and use.

```
%macro apr0qtc0conc0dose0plot(
  population_from = lptda.adsl
 ,population_where =
 ,observation_from = lptda.adeg
 ,observation_where =
 ,therapy_cd_var=trt01an
 ,therapy_des_var=trt01a
 ,rename_output =
 ,output_fileref = fptog
 ,page_orientation = portrait
 ,page_margins =
 ,font_size = 10
 ,subtitle=
 ,cutoff_dt =
 ,end_notes=
 ,data_citation_txt=
 ,debug = &gmvar_debug);
```

**Display 2: example of macro parameter naming conventions**

- Make the macro more robust
  A good macro should include checks that help prevent errors. For example, default values can be set for macro parameters if the user does not provide them.

```
%if %sysevalf(%superq(page_margin_count)^=4,boolean) %then %do;
    %put ================== Problem in macro %upcase(&sysmacroname) call ==================;
    %put The parameter "page_margins" must be 4 numbers in inch with order of top bottom left
    %put page_margins=&page_margins;
    %put The macro will not process your request.;
    %put ================== Problem in macro %upcase(&sysmacroname) call ==================;
    %goto exit;
%end;
```

**Display 3: example syntax of macro parameter value check**

- Use a consistent approach to create output files
  Some companies use ODS RTF and PROC REPORT to create tables. Others use in-house macros to generate RTF tables. Using company-approved macros to create tables, listings, and figures helps ensure that the output format meets client expectations, such as those of the clinical team or CSR author. Important formatting details include table margins, border lines, font size, title format, citation placement, and footnote placement and size. Several papers have already introduced SAS macros or R packages for RTF output. These papers include Qi and Zhang (2007), Wang et al. (2020), Zhang et al. (2021), and Chen et al. (2021).

- Use a consistent approach for titles and footnotes
  Title formatting should consider font size, bold text, and whether line breaks are soft or hard return between the title and the table body. Footnotes also need consistent formatting, such as left or center alignment, placement inside or outside the table box, and whether footnote symbols are alphabetical or symbolic. Usually, data citation should be included in the table.

```
%if %length(&title1_text) > 0 %then %let title1 = &title1_text;
%else %let title1=Model Evaluation for &paramcd_display: Quantile-quantile Plot of

%if "&subtitle" ne "" %then %do;
    %let title_out = %nrbquote(&title1 | &subtitle.);
%end;
%else %do;
    %let title_out = %nrbquote(&title1);
%end;
%let ftitle=%qsysfunc(tranwrd(&title_out, %str(|), %str(^{newline})));
```

**Display 4: example syntax of inserting a soft return between two title lines**

- Clean up intermediate datasets
  A good macro should also manage temporary datasets properly. For example, it should delete intermediate datasets at the end of the macro.

```
%if %qupcase(&debug) = %quote(N) %then
 %do;
  proc datasets library=work memtype=data nolist nowarn;
   delete _all_;
  quit ;
 %end;
```

**Display 5: example syntax of deleting intermediate datasets at the end of the macro**

- Optional: Use SAS MFILE for output capture
  The SAS MFILE option sends the output from the MPRINT option to an external file. Some company-standard macro libraries use this feature to create open code for future reruns or ASCII file generation.

```
%*** Turn on options to start outputing resolved code file if requested ***;
%if %qupcase(&debug)=%quote(ANALYSIS) %then %do;
    %let _lstpath=%sysfunc(pathname(fptol));
    filename mprint "&_lstpath./&rename_output.gen.sas" lrecl=2048;
    %let rc=%sysfunc(fdelete(mprint));
    options nomautolocdisplay mfile mprint;
%end;
```

**Display 6: example syntax of using SAS MFILE sending the output from the MPRINT to an external file**

- Graphic Macro specific:
  - Initial global graph parameters and style template

## DESIGN MACROS WITH BALANCED DATA-DRIVEN DEFAULTS AND USER-DEFINED PARAMETERS

When creating a table or a plot, an automated data-driven macro is usually preferred. In QTC analysis, a table with columns based on ECG measurement categories or a plot with the X-axis based on ECG measurement time points, the macro can identify the categories or time points and set up the output automatically.

However, this approach depends on having a well-prepared analysis dataset with the correct values and formats. For example, if the plot is divided into three panels: “Cycle 1 Day 1,” “Cycle 2 Day 1,” and “Cycle 3 Day 1,” we must decide whether to make the formatting change in the analysis dataset or inside the macro. If the formatting is done in the macro, it must also be flexible enough to handle future variations.

In some cases, user-defined parameters are a better choice because they simplify the macro and make it more robust. The programmer can use these parameters to control how the X-axis labels are displayed. The example below shows how user-defined parameters can be used to select panels and time points, define display values on the plot, and parse those values into multiple macro parameters for greater flexibility.

```
,panel_variable         = %str(visit)
,panel_value            = %str(Cycle 1 Day 1|Cycle 2|Cycle 3)
,panel_display          = %str(Cycle 1 Day 1|Cycle 2 Day 1|Cycle 3 Day 1)
,timepoint_variale      = %str(egtpt)
,timepoint_value        = %str(DAY 1 PREDOSE|DAY 1 2 HOUR POST DOSE|DAY 1 4 HOUR PO
,timepoint_display      = %str(Predose|2|4|6)
```

**Display 7: example of user-defined parameters for panel and time point selection**

```
%*** parse &panel_value;
%do j=1 %to %eval(%qsysfunc(count(&panel_value.,|))+1);
    %let panel_&j.="%qscan(&panel_value.,&j.,|)";
    %let paneltot=&j.;
%end;

%*** parse &panel_display;
%do j=1 %to %eval(%qsysfunc(count(&panel_display.,|))+1);
    %let panel_d&j.="%qscan(&panel_display.,&j.,|)";
%end;
```

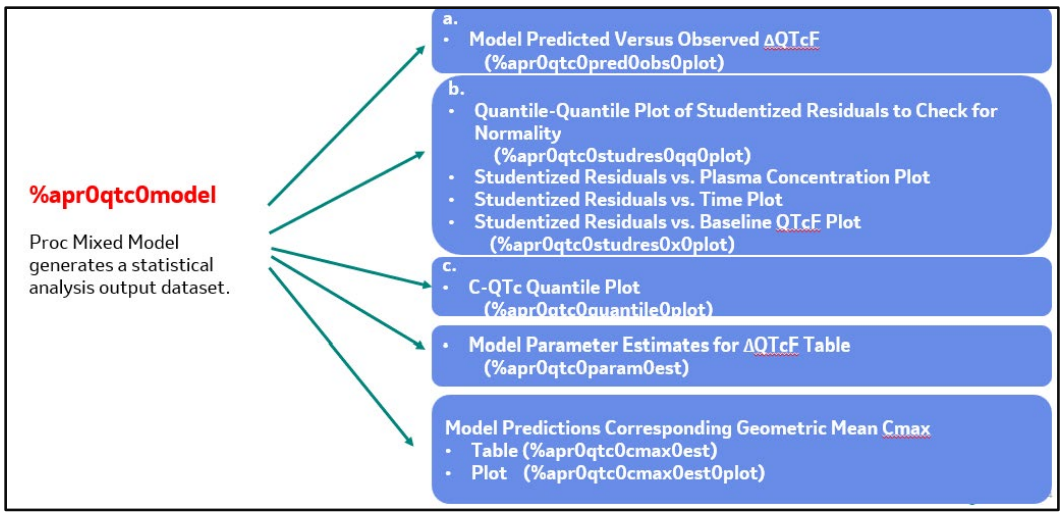
**Display 8: example of how to parse the user-defined parameter value into multiple macro parameters**

## DESIGNING A CLEAR SAS MACRO OR R FUNCTION LIBRARY ARCHITECTURE

In addition to the standard format of each individual macro, a SAS macro or R function library needs a clear design. Statisticians and programmers need to know which macro to use for each analysis purpose. To support this, the library includes macros for each type of table and plot.

| Category | Analysis Objective | Deliverable Description | Macro(s) |
|---|---|---|---|
| **Assumption Checking** | No drug effect on HR | Time-panel plot of QTc vs time by treatment | %apr_qtc_time_panel_plot |
| | QTc interval is independent of HR | Regression plot and table of QTc vs RR interval | %apr_qtc_rr_reg_plot<br>%apr_qtc_rr_reg_table |
| | No time delay between drug concentrations and ΔQTc | Concentration vs ΔQTc plot by dose | %apr_qtc_conc_dose_plot |
| | Linear C–QTc relationship | Concentration vs ΔQTc time-panel plot | %apr_qtc_conc_time_panel_plot |
| **Model Evaluation** | Predicted vs observed model fit | Scatter plot with LOESS smooth and 90% CI | %apr_qtc_pred_obs_plot |
| | Residual diagnostics: normality | Q–Q plot of studentized residuals | %apr_qtc_studres_qq_plot |
| | Residual diagnostics: vs concentration | Residual vs concentration plot with LOESS and 90% CI | %apr_qtc_studres_x_plot |
| | Residual diagnostics: vs time | Residual vs time plot with LOESS and 90% CI | |
| | Residual diagnostics: vs baseline QTcF | Residual vs baseline QTcF plot with LOESS and 90% CI | |
| | Distribution assessment | C-QTc quantile plot | %apr_qtc_quantile_plot |
| **Model-Based Predictions** | Estimate treatment effect on QTc | Parameter estimates for ΔQTcF | %apr_qtc_param_est |
| | Exposure–response prediction | Predicted ΔQTcF at geometric mean Cmax (with 90% CI) | %apr_qtc_cmax_est_plot |
| **Secondary Analyses** | Descriptive analysis | Time-panel plots and summary statistics | %apr_qtc_time_panel_plot, %apr_qtc_chg |
| | Categorical analysis | QTc categorical analysis table | %apr_qtc_category |

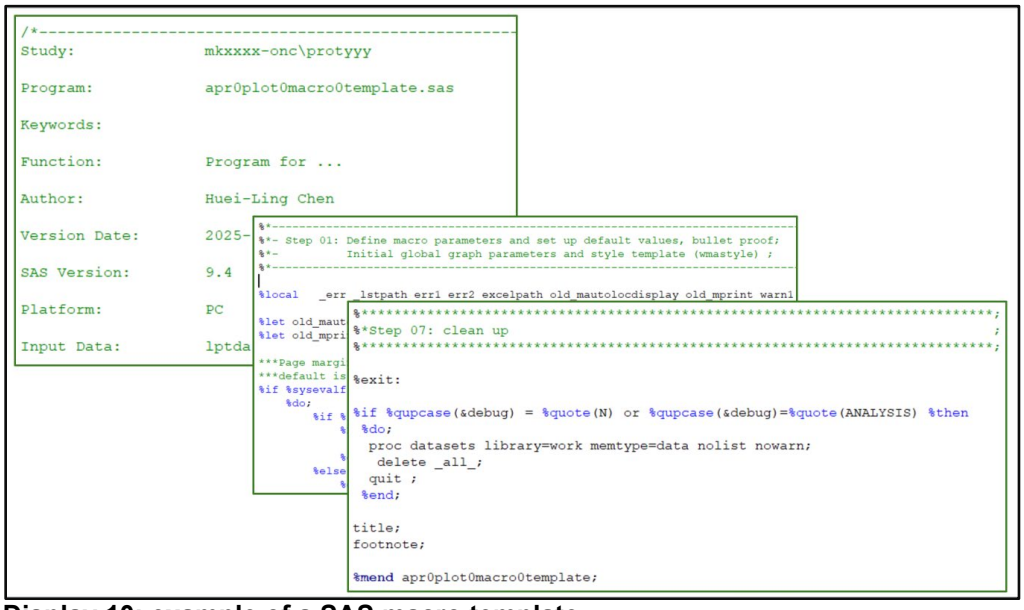
Note that the plots and tables in the two sections on **model evaluation** and **model-based predictions** need to come from a single statistical source: the PROC MIXED model. It is important to ensure that the statistician leads the statistical procedure section. In the chart below, the macro **%apr0qtc0model** runs the PROC MIXED model and generates a statistical analysis output dataset. The table and plot macros then read these output datasets to create the tables and plots. A poor design for the macro library is to repeat the statistical procedure in each output macro, because this is difficult to maintain and can lead to conflicting results.




**Display 9: flow from a central statistical procedure macro to output macros**

## DEVELOP AND PROVIDE STANDARDIZED SAS MACRO OR R FUNCTION TEMPLATE FOR TABLE AND FIGURE

Good macro templates should be prepared for different target deliverables, such as tables and figures. Team members can learn the standard macro structure by using these templates when developing macros.



```
/*-----------------------------------------------------------
Study:              mkxxxx-onc\protyyy

Program:            apr0plot0macro0template.sas

Keywords:

Function:           Program for ...

Author:             Huei-Ling Chen

Version Date:       2025-

SAS Version:        9.4

Platform:           PC

Input Data:         lptda
```

```
%*-----------------------------------------------------------------------
%*- Step 01: Define macro parameters and set up default values, bullet proof;
%*-          Initial global graph parameters and style template (wmastyle) ;
%*-----------------------------------------------------------------------
%local  _err _lstpath err1 err2 excelpath old_mautolocdisplay old_mprint warn1
```

```
%*************************************************************************;
%*Step 07: clean up                                                      ;
%*************************************************************************;

%exit:

%if %qupcase(&debug) = %quote(N) or %qupcase(&debug)=%quote(ANALYSIS) %then
 %do;
  proc datasets library=work memtype=data nolist nowarn;
   delete _all_;
  quit ;
 %end;

title;
footnote;

%mend apr0plot0macro0template;
```

**Display 10: example of a SAS macro template**

## CONCLUSION

The C-QTc analysis relies heavily on a central model and on many plots based on model estimates and predictions. Because the output is large and the plots are often complex and customized, a good macro library is needed to make programming easier. The library should be built with a clear design and should be easy to use in other studies. As multilingual programming becomes more common in the pharmaceutical industry, these ideas should apply not only to SAS macros but also to R packages built for the same purpose. New AI agents that generate SAS or R code can also follow these principles. They can use guidance as prompts to create better macros or functions. The lead programmer should provide

clear guidance and templates for the AI agent and make sure the macro architecture is well planned before programming begins.

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